

Deepak Mallampati

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Kent, OH -- F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Healthcare AI/ML engineer (2.5+ years) building production clinical-NLP and LLM extraction systems with HIPAA discipline. Track record converting unstructured clinical text into structured outputs: RAG (~35% retrieval precision gain), agentic LLM workflows (>30% hallucination reduction), and predictive ML (15-20% recall improvement on high-risk patient categories). Comfortable owning the full loop -- prompt strategies, fine-tuning, model evaluation, observability, and FastAPI/Docker deployment in HIPAA-conscious environments. Independently shipped **Manga Lens** (live Chrome extension, 4 AI providers) and **Agentic Healthcare Claims** pipeline with schema-validated multi-agent contracts.

CORE COMPETENCIES

Clinical NLP & LLM Extraction . Prompt Engineering & Fine-Tuning . RAG & Vector Search . PyTorch / Hugging Face / Transformers . Python / FastAPI / Flask . Model Evaluation & Benchmarking . HIPAA & De-identification . EHR Data Pipelines . Docker / Observability . Information Extraction & Entity Resolution . Production ML Deployment

EXPERIENCE

Progress Solutions Inc.

Jul 2025 -- Present

Jr. AI/ML Engineer Trainee -- USA | Healthcare Technology, AI Consulting, SaaS Healthcare

- Built and deployed **RAG systems** for clinical knowledge retrieval over EHR documentation; recursive semantic chunking + transformer-based embeddings improved retrieval precision by ~35% and reduced irrelevant results by >30%; response alignment exceeded 90% in evaluation.
- Developed **agentic LLM workflows** for multi-step clinical queries (eligibility, care navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~25%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show, care engagement, and support prioritization; improved recall by 15-20% on high-risk patient categories while holding precision >90% via class weighting + threshold calibration.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services containerized with Docker; structured logging + load simulation reduced post-deployment defects by ~30%.
- Built EHR preprocessing pipelines (Pandas, NumPy) for appointment histories, support tickets, and clinical extracts; raised dataset reliability above 98% and cut downstream model instability by ~40%.
- Maintained HIPAA-conscious data governance: de-identification, data lineage docs, evaluation audit trails, and stakeholder-facing system-limitation documentation.

Suvidha Foundation

Jan 2022 -- Mar 2022

Machine Learning Engineer -- Hyderabad, India | Nonprofit Tech, Video Analytics, Applied NLP

- Built **transformer-based video summarization** for 5,000+ recorded sessions; transcript preprocessing + hierarchical summarization reduced manual review time by 60-70%.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~30% and raised contextual answer accuracy >85%.

Energy Solutions International (Emerson)

Jun 2022 -- Apr 2023

Database & DevOps Performance Engineer (Intern) -- Hyderabad, India | Oil & Gas Enterprise ERP

- Optimized SQL/T-SQL stored procedures for an order-to-cash platform; reduced query execution ~20% and data retrieval latency ~25%.
- Built CI/CD pipelines with Jenkins; reduced deployment errors >30% and improved release cycle ~35-40%.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake -> Validation -> Consistency Analysis -> Duplicate Detection -> Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN similarity for duplicate claim detection; explainable risk scoring with audit-ready reasoning traces.

Python, LangChain, vector search, FastAPI, Docker, healthcare ontology

Manga Lens shipped -- Chrome Web Store

Solo-built Chrome extension integrating **4 AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local) with provider-agnostic abstraction. Multi-section capture pipeline with viewport-slice screenshots, coordinate remap, 7-day translation cache, 29 site selector configs. Full production discipline: privacy policy, scoped host permissions, Chrome review.

TypeScript, Manifest V3, Claude/GPT-4o/Ollama, content scripts + service workers

Dream Decoder -- AI Multimodal Platform

Full-stack FastAPI + React/TS app with coordinated multimodal APIs (Perplexity Sonar, Replicate). **Intermediate structured prompt transformation layers** improved contextual alignment ~**30%** and first-pass image success ~**25-30%** over naive direct prompting.

FastAPI, React, TypeScript, Perplexity Sonar, Replicate

Driver Drowsiness Detection -- Real-Time YOLOv8

Replaced two-stage CNN with unified **YOLOv8** detection-and-classification; reduced inference latency ~30%; sliding-window confidence aggregation cut blink-driven false positives ~25%.

YOLOv8, OpenCV, video pipeline

EDUCATION

Master's Degree -- Kent State University, Ohio, USA

Aug 2023 -- May 2025

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (PostgreSQL, T-SQL), TypeScript, React

LLM & GenAI: RAG, agentic workflows, prompt engineering, fine-tuning, structured outputs, evaluation pipelines, guardrails, grounding, embeddings, vector search

ML Libraries: PyTorch, Hugging Face Transformers, scikit-learn, XGBoost, LangChain, LlamaIndex, spaCy (familiar)

Computer Vision & Multimodal: YOLOv8, OpenCV, Stable Diffusion, ControlNet

Infra & DevOps: Docker, CI/CD (Jenkins), Grafana, REST APIs, MLOps/LLMOps, observability/logging

Domains: Healthcare AI (HIPAA-conscious), clinical NLP, EHR data, claims processing, Applied AI, Workflow Automation

Work Authorization: F-1 OPT -- authorized to work in the United States now; future sponsorship required